

Report Exam Computational Finance: "P + L".

Mengoni Gabriele, ID: 0001123008.

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Contents

1	Introduction: problem setting.	1
2	Heston Model.	1
2.1	MC Euler Scheme.	1
2.2	QE Scheme.	2
2.3	Fourier Methods.	3
3	Calculating the Prices.	3
4	Implied Volatility.	4
5	Profit and Loss.	5
6	VaR.	5
7	Conclusion.	6
8	Appendix A.	7
9	Appendix B.	7
10	Appendix C.	8
11	Appendix D.	9
12	Appendix E.	11
13	Appendix F.	12
14	Appendix Z.	13

1 Introduction: problem setting.

We are asked to solve 2 macro exercises, the first one is to model the evolution of an asset according to the Heston model, a stochastic model that adds "stochasticity" also to the volatility itself and then to find the price of an option. The second one asks us to find the PnL given that we are unaware of the real behavior of the market and so we rely on Black-Scholes, then we have to perform the same task assuming some shifts on the implied volatility.

One **fundamental** task will be to ensure that our methods are correct, and we did this by computing some backup techniques to use as comparison methods.

2 Heston Model.

We briefly present the Heston model, which is a stochastic volatility model where the volatility follows a CIR model.

$$\begin{aligned} S_t &= S_0 e^{X_t} \\ dX_t &= -\frac{\nu_t}{2} dt + \sqrt{\nu_t} dW_t, \quad X_0 = 0 \\ d\nu_t &= \lambda(\bar{\nu} - \nu) dt + \eta \sqrt{\nu_t} dY_t, \quad \nu_0 = \sigma^2. \end{aligned}$$

And the two Wiener process differentials are correlated:

$$\mathbb{E}[dW_t dY_t] = \rho dt$$

The set of parameters we are given are reported in **Appendix A**.

We started our computations to solve the following equation:

$$\Pi(0) = P(0, T) \mathbb{E}[(k - S(T))^+]$$

Where $P(0, T)$ is the discounting process, $\mathbb{E}[(k - S(T))^+]$ stands for the payoff of the put option, and $(x)^+$ is the "positive-part" function – that is $\max(x, 0)$ – from now on we will refer to the put payoff as $\psi(x)$, since we are dealing with only one option terminal condition, we do not risk getting confused. The other parameters necessary to price the option are also given in **Appendix A**.

2.1 MC Euler Scheme.

Our first approach was to use the simple Euler scheme as proposed in [Professor Pietro Rossi, lecture notes 2025]¹ that reads:

$$\begin{aligned} X_{n+1} &= X_n + \left(r - \frac{\nu_n}{2}\right) \Delta t + \sqrt{\nu_n \Delta t} (\rho \xi_Y + \sqrt{1 - \rho^2} \xi_Z) \\ \hat{\nu}_{n+1} &= \nu_n + \lambda(\bar{\nu} - \nu_n) \Delta t + \eta \sqrt{\nu_n \Delta t} \xi_Y \\ \nu_n &= (\hat{\nu}_n)^+. \end{aligned}$$

Where $X_n := X_{t_n}$, $\nu_n := \nu_{t_n}$ and $\xi_Y, \xi_Z \sim N(0, 1)$. Here we encountered the first problem, that is the martingality assumption was not satisfied, in fact $\mathbb{E}[S_t] \neq S_0 \quad \forall t$, (the computational method to test this assumption is reported in **Appendix B**.) this was investigated with the plot given in **Appendix B**. Moreover we noted that the Feller condition is not respected, that is $1 < \frac{\eta^2}{2\lambda\bar{\nu}} \approx 4.4$, this means that the CIR variance process could reach 0, making our dS_t being driven only by the $-dt$ term. Because of this we switched to a Quadratic Exponential (QE) scheme.

¹Rossi, 2025, pg 87.

2.2 QE Scheme.

We actually used 2 different discretization, both utilizes the QE scheme for the variance processes, as proposed by [Anderson, 2007]² but in the first the S_t trajectories are computed adapting the function given during the lectures, instead the second uses [Anderson, 2008]³ to discretize the asset. We also implement a trick, since the volatilities paths are very expensive in terms of computations, to simulate the n path we simulate just v volatilities, and then s paths are simulated for each volatilities, giving $n = vs$, from now on we set $n = 2'000'000$ and $v = 2'000$, moreover we took a $dt = 1/252$ and so we have $\text{int}(252 \cdot 1.13)$ time steps. The volatility process, shared by both methods is obtained as in the following algorithm:

$$m = \theta + (V_{t-1} - \theta)e^{-\kappa\Delta t},$$

$$s^2 = \frac{V_{t-1}\sigma^2 e^{-\kappa\Delta t}}{\kappa}(1 - e^{-\kappa\Delta t}) + \frac{\theta\sigma^2}{2\kappa}(1 - e^{-\kappa\Delta t})$$

Those are the conditionals moments of the volatility distribution, here we adopt the classical CIR model parameter notation. Refer to **Appendix A** for their corresponding values. Then we find: $\phi = \frac{s^2}{m^2}$ and make a distinction: when $\phi \leq 1.5$ we approximate the vol. process (V_t) with a chi-squared:

$$b = \sqrt{\frac{2}{\phi} - 1 + \sqrt{\frac{2}{\phi}(\frac{2}{\phi} - 1)}},$$

$$a = \frac{m}{1 + b^2},$$

$$V_t = a(b + Z_v)^2.$$

Instead if $\phi > 1.5$, the exponential distribution is used:

$$p = \frac{\phi - 1}{\phi + 1},$$

$$\beta = \frac{1 - p}{m},$$

$$V_t = \begin{cases} 0, & U_v \leq p, \\ \frac{1}{\beta} \ln(\frac{1-p}{1-U_v}), & U_v > p \end{cases}$$

Now for S_t , in the first case we use an integral approximation as proposed in [Rossi, 2025]⁴

$$S_{n+1} = S_n \exp\left\{-\frac{1}{2}\Delta\mathcal{I}_n + \frac{\rho}{\eta}(\nu_{n+1} - \nu_n - \kappa\theta(t_{n+1} - t_n) + \kappa\Delta\mathcal{I}_n) + \sqrt{(1 - \rho^2)\Delta\mathcal{I}_n}\xi\right\},$$

$$\Delta\mathcal{I}_n = \int_0^{t_{n+1}} \nu(s)ds - \int_0^{t_n} \nu(s)ds.$$

Instead in the second case we used [Andersen, 2008]:

$$S_t = S_{t-1} \exp\{(r - q)dt + K_0 + K_1 V_{t-1} + K_2 V_t + \sqrt{K_3 V_{t-1} + K_4 V_t} Z,$$

$$K_0 = -\frac{\kappa\rho\theta}{\sigma}dt,$$

$$K_1 = \left(\frac{\kappa\rho}{\sigma} - \frac{1}{2}\right)\gamma_1 dt - \frac{\rho}{\sigma},$$

$$K_2 = \left(\frac{\kappa\rho}{\sigma} - \frac{1}{2}\right)\gamma_2 dt + \frac{\rho}{\sigma},$$

$$K_3 = (1 - \rho^2)\gamma_1 dt,$$

$$K_4 = (1 - \rho^2)\gamma_2 dt.$$

²”Efficient Simulation of the Heston Stochastic Volatility Model”, Andersen 2007

³”Simple and efficient simulation of the Heston stochastic volatility model.”, Andersen 2008

⁴Computational Finance lecture notes, Pietro Rossi 2025

Where $\gamma_{1,2}$ are some model parameters and $Z \sim N(0, 1)$. In **Appendix C** we show some plot related to the first QE scheme.

2.3 Fourier Methods.

To get a comparison on the goodness of our computed payoff, that will all be reported in the next section, we choose to implement 3 Fourier methods, the first (1) is simply the inverse Fourier transform (IFT) of a function (ψ_T) of the Heston process characteristic function(ϕ_T), the second (2) is the COS Method and finally (3) the famous [Carr & Madan, 2001]⁵ discrete Fourier transform, implemented through the FFT method. (1) For the inverse integral transform:

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \psi_T(v) dv = \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} e^{-ivk} \psi_T(v) dv,$$

$$\psi_T = \frac{e^{-rT} \phi_T(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}.$$

We approximated the integral via a quadrature method, and the second equality in the first equation holds because C_T is real implying that ψ_T is odd in its imaginary part and even in the real part, be careful here, ψ_T and ϕ_T are not the same that we defined before, in fact ϕ_T is the characteristic function of the process, to have a more deep derivation and some other insight on how we choose the characteristic function, look at **Appendix D**. instead ψ_T is a function of it and some other parameters. (2) For the COS method:

$$P_T = \frac{e^{-r(T-t)}}{b-a} Re[g_0 + \sum_{n=1}^N 2\phi_t\left(\frac{n\pi}{b-a}\right) e^{in\pi \frac{-a}{b-a}} g_n]$$

This method prices the put directly, i.e. we don't need to convert the call price in put price afterward. To get the derivations for g_n and g_0 look at **Appendix D**. (3) Finally we implement the Carr & Madan formula:

$$C_T(k_u) = \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^N \exp\{i(bv_j - \frac{2\pi}{N}(j-1)(u-1))\} \psi(v_j) \frac{\eta}{3} [3 + (-1)^j - \delta_{j-1}]$$

Where the part $\eta/3[\dots]$ is the trapezoidal rule, and δ_n is the Kronecker delta function, that is equal to 1 when $n = 0$ and 0 elsewhere. For a more thorough explanation refer to **Appendix D**. The formidable part of this last implementation is that it gives the price of the call not for only one strike price but for a grid of them making it exceptionally computationally efficient. Methods (1) and (3) gives us the price of the corresponding call that we transformed in a put, instead (2) gives us directly the put price.

3 Calculating the Prices.

Here finally we find $\Pi(0)$ in such way: for 2 Monte Carlo QE schemes we find $\mathbb{E}[(K - S_T)^+]$ starting by applying the payoff function (ψ) – defined in the first paragraph of chapter 2 – and then we compute the average of the array, that is, the array that refers to time T:

$$\Pi(0)e^{rT} = \frac{\sum_{j=1}^J \psi(S_T^j)}{J}$$

Then we just discount by multiplying both side by the discount process that is $P(t, T) = e^{-r(T-t)} \implies P(0, T) = e^{-rT}$, and then the final price will be $\Pi(0) \pm MC_{error}$, where the error

⁵"Option valuation using the FFT", Carr & Madan, 2001

is computed as explained in **Appendix B**, with the "lean" scheme variation for the volatility paths, and by taking the std of $\psi(S_T^j)$.

For the Fourier methods (1) and (2) we transformed call price in put price with put-call parity and got the option prices at $t=0$, we give them in the following table:

Method	Put Price	Absolute Difference	Relative Error (%)	Execution Time (s)
Integral Fourier	0.097539	0.000280	0.286433	0.004622
COS Method	0.097539	0.000280	0.286433	0.031792
Carr & Madan FFT	0.090246	0.007573	7.742053	0.098722
Different QE Scheme	0.096237 $\pm$ 0.0015	0.001583	1.618003	17.950890
Our Heston Model	0.097819 $\pm$ 0.0081	0.000000	0.000000	38.599062

Table 1: Comparison of different methods for put price calculation

Our QE scheme – that is the one with the `mc_heston` function adapted from the code given in the lecture notes is set to be our reference point – we can see that the percentage errors, relative to the other methods range from 0.28% to 7.74%; the first relate to the Integral IFT and the second to the Carr & Madan implementation. This is to be expected since as said before the Carr & Madan gives us grid of strike price, so when we recover the price we do that according to the strike, on the grid, that minimizes the distance from the one we want, i.e.: $C(k_u) | \arg \min_u (|k_u^2 - k^2|)$. The table also add a column to compare the execution time for the pricing methodologies, our QE scheme is the slowest and the Carr & Madan formula is the slowest Fourier methods but while the integral and the COS price only 1 strike the Carr formula prices for strikes in $[-\frac{\pi}{0.25}, \frac{\pi}{0.25}]$, $\frac{\pi}{0.25} \approx 12.56$ and the density of the grid is defined by $\frac{2\pi}{0.25 \cdot 4096} \approx 0.00613$ that is the spacing between the 4096 strikes priced, even if not all of them are valid since some are in $\mathbb{R}^-$. We can conclude that our result makes sense, since they are all very close and inside the Monte Carlo errors, thus following with the exercise.

4 Implied Volatility.

For the second part was fundamental to find the implied volatility and the implied volatility surface, the goodness of our work was easy to check in this case since we had a benchmark file that we used to validate our results, the comparison is given in **Appendix E**. So from now on we will assume that our findings are in fact accurate. The implied volatility is computed by adjusting the σ_{BS} in order to make the BS price equal to the one given in the market. We did this with a standard bisection function, in which we feed in the BS price and the market price (Heston process) for various time to maturity and strikes, we then plotted the following:

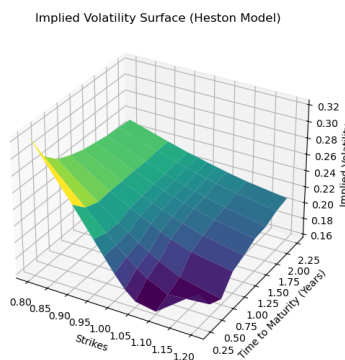


Figure 1: Implied volatility surface.

From our parameters we can deduce that the graph should converge to $\sigma_{IV} = \sqrt{\bar{v}} \approx 0.24$ as $\tau \rightarrow \infty$ and that is the case. This is because of the following: it is a well known fact that

$\sigma_{BS}^2 \approx \frac{1}{T} \int_0^T \mathbb{E}[v_u | \mathcal{F}_t] du$ see for example [Gatheral, 2006]⁶ and [Comte, 1998]⁷, then it is also possible to show that the expected value of the CIR process is $\mathbb{E}[\nu_t] = \nu_0 e^{-\lambda t} + \bar{\nu}(1 - e^{-\lambda t})$ so if we take the integral and then we set $T \rightarrow \infty$ all the terms vanish expect for $\bar{\nu}$ yielding the result: $\sigma_{BS} \underset{T \rightarrow \infty}{\sim} \sqrt{\bar{\nu}}$, thus proving our claim. To do an additional brief sanity check of the plotting/computation method for the surface we just calculated the implied volatility smile for $t=1.13$:

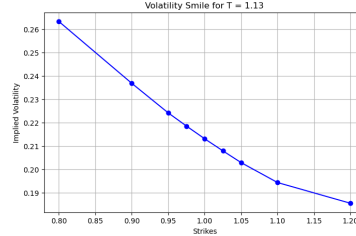


Figure 2: Implied volatility smile for $t=1.13$.

The plot correspond in fact to the "slice" of the surface when time to maturity is 1.13.

5 Profit and Loss.

We reached the last part of the exam, where we are asked to compute the random variable defined as:

$$\Upsilon := \Pi(0) - P(0, t_m)\Pi(t_m)$$

where :

$$\Pi(t_m) = P(t_m, T)\mathbb{E}[(K - S_T)^+ | \mathcal{F}_{t_m}]$$

Here we assume that we are at $t_m = 6$ months, completely unaware that the market follows the Heston model and so we find $\Pi(t_m)$ using BS and σ_{IV} , we did that by taking all the simulations in our QE scheme at the step ($t = \frac{252}{2} = 126$), that corresponds to the 6 month mark:

$$\Pi^j(t_m) = BS_{put}(S_{0=t_m}^j, K, \sigma_{IV}, T - t_m, r)$$

So we have j possible outcome given by the BS put price of the j simulated paths. Each of them is additionally discounted to $t = 0$ with $P(0, t_m)$, we can now compute the variable Υ . We have to notice that this variable –that indicates our PnL in case we bought the put at time 0, and it evolved to t_m – is actually inverted, in fact when the option has an higher value at t_m than at $t = 0$, Υ is negative, but in reality we would have seen a profit. This will have some implications in the VaR calculations. We are asked to plot the graph originating from Υ , and we also have to do the same assuming some shifts in the σ_{IV} those are $\sigma_{IV}(1 + \Delta)$, $\Delta = [\pm 0.05, \pm 0.1, \pm 0.5]$. All the PnL distributions plots are given in **Appendix F**.

6 VaR.

We are finally asked to compute the Value at Risk (VaR) with the $p = [0.01, 0.05, 0.1]$ that is the probability such that a lose of that entity or more will happen with $\mathbb{P} \leq p$. Usually:

$$VaR_p(X) = q_{-X}(1 - p)$$

⁶"The volatility surface, a practitioner's guide", Gatheral 2006

⁷"Long Memory in Continous-Time Stochastic Volatility Models", Comte, 1998

But actually our Υ is already the reflection of the PnL, remember when we would have made a profit, i.e. $P(0, t_m)\Pi(t_m) > \Pi(0)$, Υ is less than 0, our Υ is already $-X$ without the need to reflect it. So in the above $q_{-X}(u)$ is the quantile function that is the same as $F_{-X}^{-1}(u)$. We find the VaRs and give them in this table:

Shift	VaR 10%	VaR 5%	VaR 1%
No shift	0.0443	0.0535	0.0699
-50.00%	0.0960	0.0972	0.0978
-10.00%	0.0833	0.0886	0.0950
-5.00%	0.0811	0.0868	0.0941
5.00%	0.0765	0.0830	0.0920
10.00%	0.0742	0.0810	0.0908
50.00%	0.0535	0.0624	0.0775

Table 2: Value at Risk (VaR) for Different Shifts

We also plot the PnL with the VaR value in **Appendix F**.

7 Conclusion.

In this report, we solved two macro exercises regarding the Heston model and option pricing and then a Profit and Loss (PnL) analysis under the assumption that the market behavior is unknown and traders rely on the Black-Scholes model. One of the fundamental aspects of our approach was to ensure that the outcomes were accurate and reliable. In order to confirm this, we employed various validation techniques, including alternative numerical methods and Fourier-based pricing approaches, to cross-validate our results. This helped us corroborate our Monte Carlo simulations and the pricings employed.

In computing the implied volatility, we also reinforced our confidence in the approach by cross-verifying our results against a given benchmark dataset. The strong correlation between our computed values and the benchmark not only confirmed the accuracy of our approach but also the robustness of our numerical techniques. This cross-verification played a crucial role in ensuring that the subsequent PnL and Value at Risk (VaR) computations were founded on sound assumptions.

Finally, our PnL analysis and VaR computations gave us a glimpse of the economic implications of applying Black-Scholes pricing under a Heston-driven regime. Stress-testing our findings against different implied volatility shifts provided us with a clear grasp of how such assumptions could influence risk estimation. Overall, the blending of multiple pricing techniques, benchmark testing, and stress-testing provided our results with strong foundation, validating our methodology and our findings so much more accurately. Some additional final remarks are given in **Appendix Z**.

8 Appendix A.

Parameter	Value
λ (κ)	7.7648
$\bar{\nu}$ (θ)	0.0601
η (σ)	2.017
ν_0	0.0475
ρ	-0.6952

Table 3: Parameter values for the Heston risk-neutral martingale measure. In the code the parameter are called as ().

Parameter	Value
k	1.03
T	1.13
$S(t_0)$	1.0
r	0.01

Table 4: Option pricing parameters.

9 Appendix B.

How to test for martingality: $\forall t \quad S_0 \in \bar{S}'_t \pm \epsilon_t$, where ϵ_t is the Monte Carlo error at time t computed as: $3 \frac{\sigma(S'_t)}{\sqrt{J}}$, where S'_t is the array of J paths for the time step t and $\bar{S}'$ is its mean, we need to be cautious when we use the "lean" scheme for volatilities, when we build v volatilities path and on them s asset paths, because in that case we need $m^j := \frac{1}{s} \sum_{k=1}^s S_k^j$, and then we use the m^j that will have a numerosity of v to compute the error the usual way.

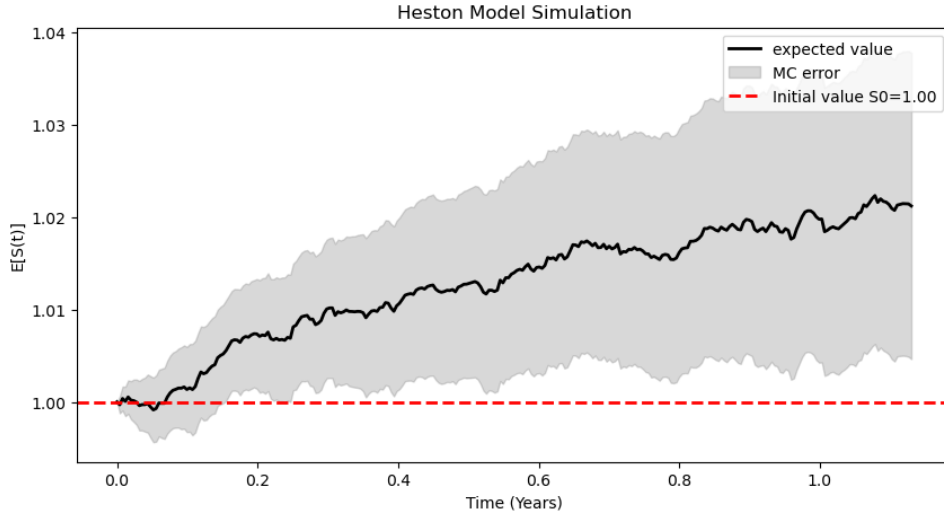


Figure 3: Martingality violation with the simple Euler.

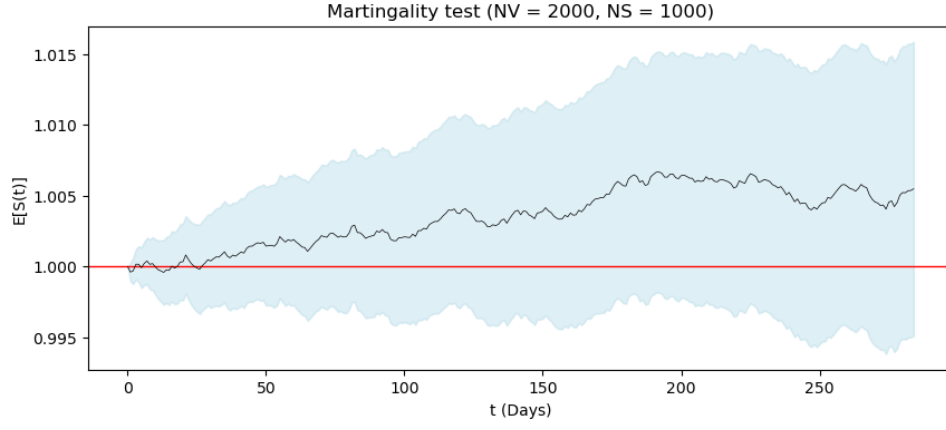


Figure 4: Martingality with QE Scheme.

10 Appendix C.

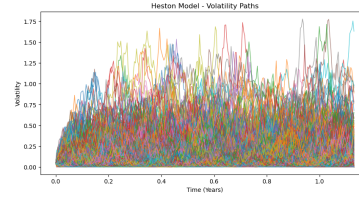


Figure 5: Volatility Paths.

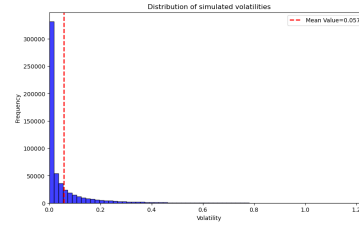


Figure 6: Distribution of the Volatilities.

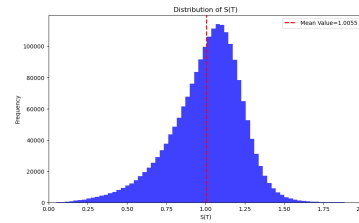


Figure 7: Distribution of S'_t .

11 Appendix D.

We had to choose between 2 characteristic function, the first proposed by [Heston, 1993] and the second proposed in [Schoutens, 2004]:

$$\begin{aligned}\phi_t(u) &= \exp\{iu\mu t + \frac{\kappa\theta}{\sigma^2} \left((\xi + d)t - 2 \ln \left(\frac{1 - g_1 \exp(d \cdot t)}{1 - g_1} \right) \right) + \frac{v_0}{\sigma^2} (\xi + d) \frac{1 - \exp(d \cdot t)}{1 - g_1 \exp(d \cdot t)} \}, \\ g_1 &= \frac{\xi + d}{\xi - d}, \\ d &= \sqrt{\xi^2 + \sigma^2(u^2 + iu)}, \\ \xi &= \kappa - \sigma\rho ui\end{aligned}$$

and the second that was:

$$\begin{aligned}\phi_t(u) &= \exp\{1ju\mu t + \frac{\kappa\theta}{\sigma^2} \left((\xi - d)t - 2 \ln \left(\frac{1 - g_2 \exp(-d \cdot t)}{1 - g_2} \right) \right) + \frac{v_0}{\sigma^2} (\xi - d) \frac{1 - \exp(-d \cdot t)}{1 - g_2 \exp(-d \cdot t)} \}, \\ g_2 &= \frac{1}{g_1}, \\ g_1 &= \frac{\xi + d}{\xi - d}, \\ d &= \sqrt{\xi^2 + \sigma^2(u^2 + iu)}, \\ \xi &= \kappa - \sigma\rho ui.\end{aligned}$$

The chose became trivial once one plots the characteristic function:

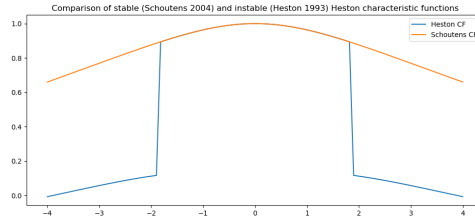


Figure 8: Plot of ϕ 's.

It is clear that the version proposed by Heston as a branch cut for some value, making the computations unstable.

(2) COS Method derivations:

Derivation for g_0 and g_n :

$$\begin{aligned}g_n &= K(\ln K - a - 1) + e^a, \quad n = 0, \\ g_n &= \frac{e^a - \frac{K}{h_n} \sin(h_n(a - \ln K)) - K \cos(h_n(a - \ln K))}{1 + h_n^2}, \quad n > 0, \\ h_n &= \frac{n\pi}{b - a}\end{aligned}$$

We now only need the $[a, b]$ and the cumulants, we get them from [Bormetti et Al, "A Generalized Fourier Transform Approach to Risk Measures", 2012]:

$$\begin{aligned}
[a, b] &= [c_1 - 24\sqrt{|c_2|}, c_1 + 24\sqrt{|c_2|}], \\
c_1 &= \ln(S_0) + rT + (1 - e^{-\kappa T}) \frac{\theta - \nu_0}{2\kappa} - \frac{T\theta}{2}, \\
c_2 &= \frac{\theta}{8\kappa^3} [-\zeta^2 e^{-2\kappa T} + 4\zeta e^{-\kappa T} (\zeta - 2\kappa\rho) + 2T\kappa(4\kappa^2 + \zeta^2 - 4\kappa\rho\zeta) + \zeta(8\kappa\rho - 3\zeta)]
\end{aligned}$$

(3) Carr & Madan:

$$\begin{aligned}
k_u &= -b + \lambda(u - 1), \quad for \quad u = 1, 2, \dots, N \\
b &= \frac{N\lambda}{2}, \\
v_j &= \eta(j - 1)
\end{aligned}$$

Here λ and η are NOT the same parameter used in the Heston simulation and N should be a power of 2, we set ($N = 2^{12}, \eta = 0.25$). k_u gives us a strike grid that ranges from -b to b.

12 Appendix E.

T	log_K	Strike	Benchmark_IV	Benchmark_Put	Computed_Put	Error_Put	Computed_IV	Put_Check
0.0806	-0.2231	0.800	0.3893	0.0008	0.000592	0.000272	0.373516	True
0.0806	-0.1054	0.900	0.3018	0.0043	0.003886	0.000917	0.295205	True
0.0806	-0.0513	0.950	0.2496	0.0095	0.008997	0.001506	0.246140	True
0.0806	-0.0253	0.975	0.2204	0.0141	0.013655	0.001873	0.218345	True
0.0806	0.0000	1.000	0.1898	0.0215	0.021034	0.002258	0.189341	True
0.0806	0.0247	1.025	0.1636	0.0338	0.033456	0.002571	0.165413	True
0.0806	0.0488	1.050	0.1531	0.0529	0.052495	0.002705	0.157920	True
0.0806	0.0953	1.100	0.1630	0.1003	0.099886	0.002726	0.184023	True
0.0806	0.1823	1.200	0.1949	0.2000	0.199464	0.002724	0.290371	True
0.1667	-0.2231	0.800	0.3520	0.0033	0.002810	0.000923	0.341611	True
0.1667	-0.1054	0.900	0.2742	0.0099	0.009154	0.001857	0.269585	True
0.1667	-0.0513	0.950	0.2316	0.0170	0.016252	0.002516	0.229342	True
0.1667	-0.0253	0.975	0.2095	0.0226	0.021814	0.002883	0.208193	True
0.1667	0.0000	1.000	0.1876	0.0305	0.029646	0.003261	0.187286	True
0.1667	0.0247	1.025	0.1680	0.0420	0.041012	0.003583	0.168917	True
0.1667	0.0488	1.050	0.1549	0.0582	0.057228	0.003801	0.158162	True
0.1667	0.0953	1.100	0.1524	0.1018	0.100656	0.003898	0.162512	True
0.1667	0.1823	1.200	0.1752	0.2001	0.198753	0.003871	0.217861	True
0.2500	-0.2231	0.800	0.3279	0.0058	0.005255	0.001521	0.322275	True
0.2500	-0.1054	0.900	0.2602	0.0146	0.013743	0.002579	0.257863	True
0.2500	-0.0513	0.950	0.2249	0.0232	0.022189	0.003266	0.223783	True
0.2500	-0.0253	0.975	0.2071	0.0295	0.028296	0.003642	0.206199	True
0.2500	0.0000	1.000	0.1898	0.0378	0.036447	0.004004	0.189220	True
0.2500	0.0247	1.025	0.1739	0.0490	0.047597	0.004353	0.174839	True
0.2500	0.0488	1.050	0.1613	0.0638	0.062245	0.004608	0.163358	True
0.2500	0.0953	1.100	0.1512	0.1039	0.102218	0.004805	0.158766	True
0.2500	0.1823	1.200	0.1637	0.2004	0.198511	0.004819	0.198525	True
0.5000	-0.2231	0.800	0.2915	0.0131	0.012005	0.002731	0.288558	True
0.5000	-0.1054	0.900	0.2439	0.0270	0.025193	0.003999	0.241427	True
0.5000	-0.0513	0.950	0.2205	0.0388	0.036459	0.004727	0.218586	True
0.5000	-0.0253	0.975	0.2091	0.0465	0.043893	0.005110	0.207468	True
0.5000	0.0000	1.000	0.1980	0.0558	0.052847	0.005482	0.196713	True
0.5000	0.0247	1.025	0.1875	0.0670	0.063787	0.005837	0.187045	True
0.5000	0.0488	1.050	0.1780	0.0802	0.076763	0.006153	0.178041	True
0.5000	0.0953	1.100	0.1633	0.1139	0.110190	0.006606	0.166500	True
0.5000	0.1823	1.200	0.1548	0.2024	0.198167	0.006833	0.170730	True
1.0000	-0.2231	0.800	0.2662	0.0266	0.024905	0.004279	0.266926	True
1.0000	-0.1054	0.900	0.2362	0.0480	0.044979	0.005764	0.236895	True
1.0000	-0.0513	0.950	0.2220	0.0635	0.059830	0.006555	0.223541	True
1.0000	-0.0253	0.975	0.2152	0.0727	0.068711	0.006958	0.217136	True
1.0000	0.0000	1.000	0.2085	0.0830	0.078689	0.007353	0.210912	True
1.0000	0.0247	1.025	0.2022	0.0946	0.090142	0.007730	0.205716	True
1.0000	0.0488	1.050	0.1961	0.1075	0.102407	0.008071	0.199704	True
1.0000	0.0953	1.100	0.1850	0.1374	0.131875	0.008709	0.191274	True
1.0000	0.1823	1.200	0.1686	0.2131	0.206370	0.009483	0.182332	True
1.5000	-0.2231	0.800	0.2563	0.0391	0.036307	0.005155	0.258044	True
1.5000	-0.1054	0.900	0.2343	0.0655	0.061324	0.006768	0.237076	True
1.5000	-0.0513	0.950	0.2241	0.0832	0.078183	0.007604	0.227529	True
1.5000	-0.0253	0.975	0.2191	0.0934	0.088115	0.008029	0.223541	True
1.5000	0.0000	1.000	0.2144	0.1044	0.098492	0.008420	0.218768	True
1.5000	0.0247	1.025	0.2097	0.1164	0.110331	0.008829	0.215263	True
1.5000	0.0488	1.050	0.2053	0.1294	0.122910	0.009210	0.211547	True
1.5000	0.0953	1.100	0.1969	0.1585	0.151307	0.009923	0.205353	True
1.5000	0.1823	1.200	0.1826	0.2286	0.219775	0.011001	0.196773	True
2.0000	-0.2231	0.800	0.2510	0.0507	0.046977	0.005832	0.254358	True
2.0000	-0.1054	0.900	0.2337	0.0809	0.075344	0.007518	0.237922	True
2.0000	-0.0513	0.950	0.2256	0.1002	0.093584	0.008389	0.230551	True
2.0000	-0.0253	0.975	0.2218	0.1110	0.103856	0.008828	0.227167	True
2.0000	0.0000	1.000	0.2180	0.1225	0.114904	0.009245	0.223904	True
2.0000	0.0247	1.025	0.2144	0.1349	0.126834	0.009663	0.221003	True
2.0000	0.0488	1.050	0.2109	0.1481	0.139407	0.010049	0.217952	True
2.0000	0.0953	1.100	0.2042	0.1770	0.167178	0.010817	0.212695	True
2.0000	0.1823	1.200	0.1923	0.2445	0.232698	0.012094	0.205384	True

Table 5: Benchmark check table.

13 Appendix F.

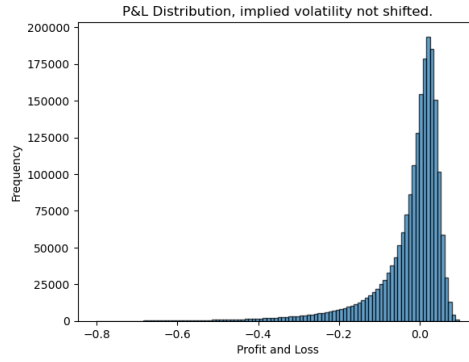


Figure 9: PnL without σ -shift.

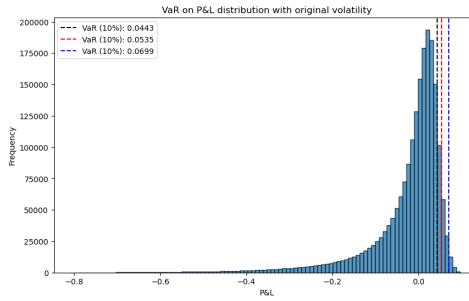


Figure 10: PnL without σ -shift and VaR.

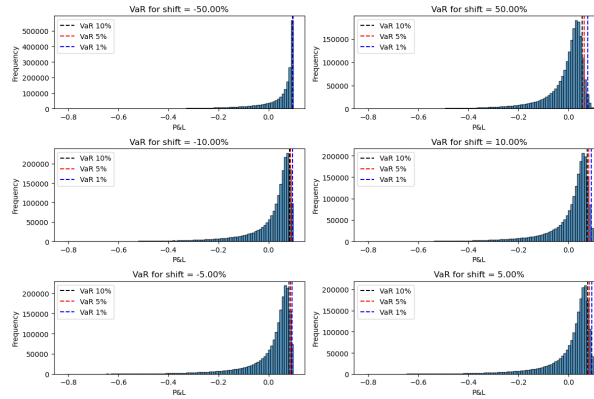


Figure 11: PnL with σ -shift and VaR.

14 Appendix Z.

This work was the "icing on the cake" of the typical QF student, here we could implement the techniques and their validation methods that a Quant will need to apply in her/his work career, even if this project was not aimed at computational speed, but just to validate the goodness of the methods applied. Researching, reading, understanding and then implementing papers and articles was the main development tool for this work, all the cited works, both in this report and in the code were crucial and very interesting readings. I also deeply appreciated the team work behind this paper, having a confrontation about some topics, about how to tackle a challenge is something that I really like. There is the need to state clearly that in some cases we used AI bot like Chat-GPT, this was done to speed up boring tasks, the most boring of them all being the writing of tables in latex format, it was used also to plot some of the graphs and to do general code feature checking to have some confirmation about the "rightness" of our scripts.